

Thm. Let $f: (a, b) \rightarrow \mathbb{R}$ be a monotonically increasing, and let $x \in (a, b)$.

Then, $f(x+)$ and $f(x-)$ exist. Specifically,

$$\sup_{a < t < x} f(t) = f(x-) \leq f(x) \leq f(x+) \leq \inf_{x < t < b} f(t)$$

Moreover, if $a < x < y < b$, then

$$f(x+) \leq f(y-)$$

Proof. Since f is monotonically increasing, for all $t \in (a, x)$, $f(t) \leq f(x)$.

Therefore, $\{f(t) \mid a < t < x\}$ is bounded by $f(x)$. Therefore, $\sup_{a < t < x} f(t)$ exists and $\sup_{a < t < x} f(t) \leq f(x)$.

Given $\epsilon > 0$, there exists $\delta > 0$ st

$$\sup_{a < t < x} f(t) - \epsilon < f(x - \delta) \leq \sup_{a < t < x} f(t)$$

Since f increases,

$$x - \delta \leq t < x \Rightarrow f(x - \delta) \leq f(t) \leq f(x)$$

which implies

$$\sup_{a < t < x} f(t) - f(t) < \epsilon \quad \text{for all } x - \delta \leq t < x.$$

Therefore, for any sequence $\{x_n\}$ in (a, x) , $x_n \rightarrow x$ implies $f(x_n) \rightarrow \sup_{a < t < x} f(t)$, thus

$$\sup_{a < t < x} f(t) = f(x-)$$

Meanwhile, the second assertion can be proved by:

$$f(x+) = \inf_{x < t < b} f(t) = \inf_{x < t < y} f(t) \quad \text{and} \quad f(y-) = \sup_{a < t < y} f(t) = \sup_{x < t < y} f(t)$$

Corollary. A discontinuous of monotone function is at most countable.

Proof. Let $f: (a, b) \rightarrow \mathbb{R}$ be a monotone increasing function. E be the set of discontinuity points.

Since f is monotone, $x \in E$ if and only if $f(x-) < f(x+)$. By density of rationals, given x ,

$$f(x-) < r(x) < f(x+)$$

for some rational $r(x) \in \mathbb{Q}$. And if x_1, x_2 , then

$$r(x_1) < f(x_1+) \leq f(x_2-) < r(x_2)$$

implies $r(x_1) < r(x_2)$. Therefore, there is injective map from E into $\mathbb{Q}$.

Thm. If $f: (a,b) \rightarrow \mathbb{R}$ satisfies Intermediate Value Property, then f cannot have first-kind discontinuity.

Proof. Suppose that f has first-kind discontinuity at $x \in (a,b)$.

Then, $f(x^-)$ and $f(x^+)$ exist, but $f(x^-) = f(x) = f(x^+)$ cannot be happen.

WLOG, suppose $f(x^-) \neq f(x)$. Put $\epsilon = |f(x^-) - f(x)| > 0$.

Since $f(x^-)$ exists, there exists $\delta > 0$ s.t. $t \in (x-\delta, x) \Rightarrow |f(x^-) - f(t)| < \frac{\epsilon}{3}$.

Now, if $f(x) > f(x^-)$, then

$$f(x - \frac{\delta}{2}) < f(x - \frac{\delta}{2}) + \frac{2}{3}\epsilon < f(x)$$

But there is no $c \in (x - \frac{\delta}{2}, x)$ s.t. $f(c) = f(x - \frac{\delta}{2}) + \frac{2}{3}\epsilon$, because

$|f(c) - f(x - \frac{\delta}{2})| < \frac{\epsilon}{3}$. If $f(x) < f(x^-)$, similarly contradiction.